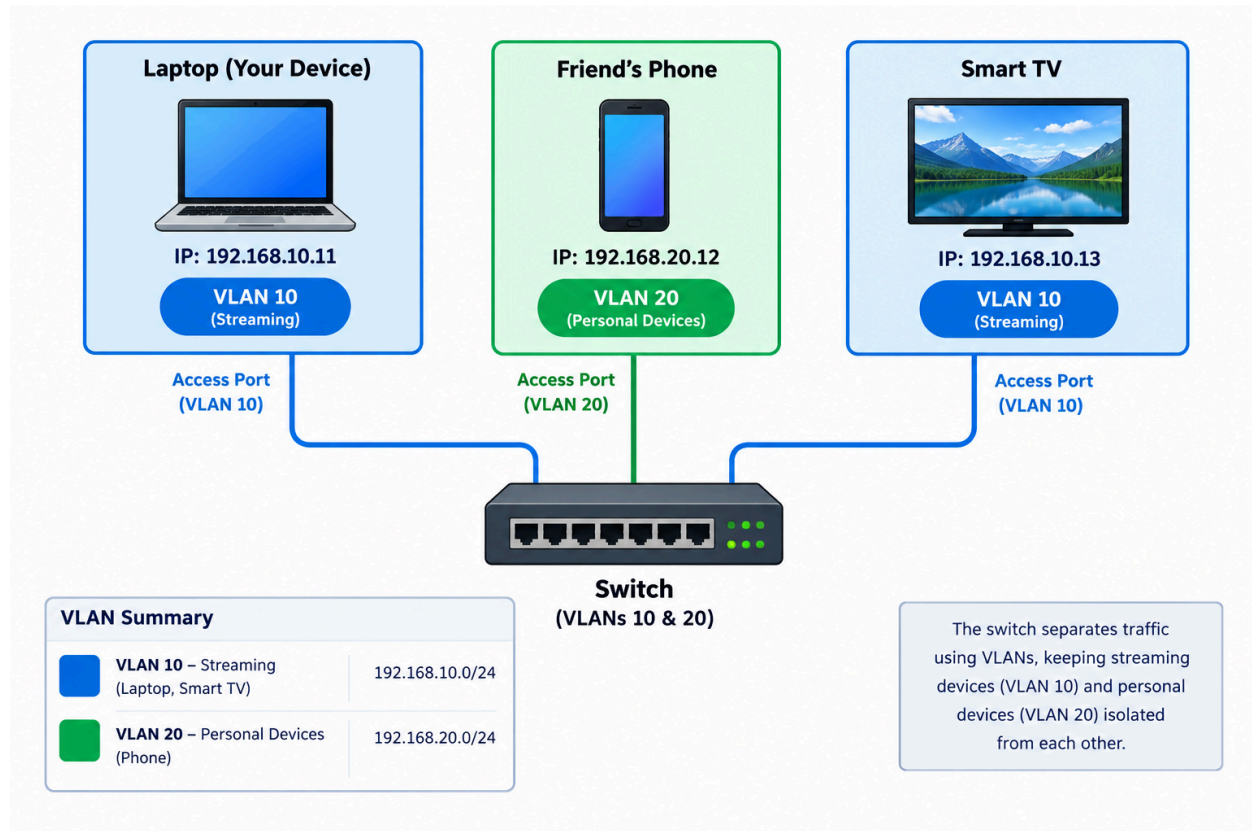


SWITCHING & VLANS

Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

ANS:



Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. **Hint:** Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

ANS:

Here are the CLI commands to configure two VLANs on a Cisco switch and assign ports to them:

1. Create the VLANs

Plaintext

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name Streaming
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Personal
Switch(config-vlan)# exit
```

2. Assign Ports to VLAN 10 (PCs 1–3)

Plaintext

```
Switch(config)# interface range fastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

3. Assign Ports to VLAN 20 (PCs 4–5)

Plaintext

```
Switch(config)# interface range fastEthernet 0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit
```

4. Verification

- Run `show vlan brief` to confirm interface assignments.
- Pings from PC 1 (VLAN 10) to PC 4 (VLAN 20) will fail because traffic between different VLANs requires a Layer 3 router (Router-on-a-Stick or Layer 3 switch).

In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

ANS:

Before Configuration (VLAN 10)

Plaintext

```
configure terminal
interface FastEthernet 0/1
switchport access vlan 10
```

After Configuration (VLAN 20)

Plaintext

configure terminal

interface FastEthernet 0/1

switchport access vlan 20

Connectivity Test

- Running a ping from the moved PC to a device in VLAN 10 now fails, while pings to VLAN 20 devices succeed.

Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

ANS:

Here is the command list using Cisco IOS CLI to automate assigning 10 ports (FastEthernet 0/1 through 0/10) to VLAN 30 using a single range command:

Plaintext

configure terminal

interface range fastEthernet 0/1 - 10

switchport mode access

switchport access vlan 30

end